SimSpace Incident Response Tool Evaluation Report

Prepared by

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Environment	SimSpace Virtual Range using Windows Hunt and Kali Hunt virtual machines
Primary tools	Wireshark, IDA Pro, Nikto, and John the Ripper
Assessment focus	Incident response support across network monitoring, malware analysis, host or web analysis, and password auditing

Executive Summary

This report presents an assessment of selected incident response and security analysis tools available through the SimSpace Virtual Range environment. The evaluation focused on four tools used across Windows and Kali Linux virtual machines: Wireshark for network monitoring, IDA Pro for malware analysis, Nikto for host and web server analysis, and John the Ripper for password auditing.

The assessment found that each tool supports a different stage of the incident response process. Wireshark provided visibility into live packet activity and suspicious communication patterns. IDA Pro supported binary and DLL inspection for malware analysis and reverse engineering. Nikto enabled rapid identification of web server misconfigurations and known weaknesses. John the Ripper demonstrated the ability to identify weak passwords and credential exposure through password auditing workflows.

Overall, the SimSpace environment proved useful for practicing incident response activities in a contained virtual setting. The evaluated tools collectively support detection, analysis, validation, and response planning functions that are relevant to real-world security operations.

Introduction

This assessment was conducted to evaluate the usefulness of selected tools in the SimSpace Virtual Range for incident response and security analysis tasks. The review focused on tools available across both Windows and Kali Linux environments and examined how each tool can assist with identifying threats, analyzing malicious behavior, and supporting remediation planning.

The tools selected for review were Wireshark and IDA Pro from the Windows Hunt environment, and Nikto and John the Ripper from Kali Linux. Together, these tools represent four important areas of incident response capability: network monitoring, malware analysis, host or web application analysis, and password security testing.

Scope

- Network packet capture and protocol analysis using Wireshark
- DLL and binary review using IDA Pro
- Host and web server analysis using Nikto
- Password auditing activity using John the Ripper
- Evaluation of how each tool supports incident response decision-making

Methodology

The assessment reviewed one tool at a time based on its operational role in an incident response workflow. In the Windows Hunt environment, Wireshark was used to inspect captured traffic and analyze packet details, while IDA Pro was used to load and inspect a DLL file for malware analysis. In the Kali Linux environment, Nikto was used to scan a target host for potential web server issues, and John the Ripper was used to review password cracking resources and password-auditing workflow steps.

The analysis focused on what each tool revealed, how it was used, and how its output could support security investigation, validation, and response planning within a simulated environment.

Assessment Snapshot

Tool	Platform	Primary use	Incident response value
Wireshark	Windows Hunt	Packet capture and protocol inspection	Supports triage of suspicious traffic, abnormal communications, and evidence review
IDA Pro	Windows Hunt	Binary and DLL analysis	Supports static analysis and reverse engineering of suspicious files
Nikto	Kali Hunt	Web server and host analysis	Supports rapid identification of exposed components, outdated services, and misconfigurations
John the Ripper	Kali Hunt	Password auditing	Supports review of weak credentials and password-related attack paths

Findings and Analysis

Wireshark for Network Monitoring

Wireshark was evaluated as a network monitoring and packet analysis tool in the Windows Hunt environment. The source lab shows packet capture from the Ethernet interface, live packet collection, protocol inspection, and review of packet properties. This demonstrates Wireshark's usefulness for observing traffic patterns, reviewing inbound and outbound communications, and validating protocol-level details during an investigation.

From an incident response perspective, Wireshark is especially useful for identifying abnormal network behavior, suspicious communications, and indicators of compromise across protocols such as TCP/IP, DNS, HTTP, SMB, ICMP, and HTTPS.

IDA Pro for Malware Analysis

IDA Pro was evaluated as a malware analysis and reverse engineering tool in the Windows environment. The source document shows the tool interface, file upload process, loading of debug information, and extracted file details from the internal database after uploading a DLL. This reflects a typical workflow for binary inspection and code analysis.

IDA Pro supports static and dynamic analysis through disassembly and debugging capabilities. In incident response, this is valuable when analysts need to determine the purpose, behavior, or potential impact of a malicious executable or DLL.

Nikto for Host and Web Server Analysis

Nikto was assessed in the Kali Linux environment as a host and web server analysis tool. The source lab includes the Nikto version output and a host-checking command against a target system. The document notes that Nikto can identify harmful files, misconfigurations, outdated server versions, and known web server vulnerabilities.

For incident response, Nikto is useful for rapidly identifying weak web server configurations and likely attack paths that may have contributed to compromise.

John the Ripper for Password Auditing

John the Ripper was reviewed as a password auditing tool in Kali Linux. The source document shows the tool interface, redirection into the John directory, directory contents, and available wordlists. This reflects a standard workflow for preparing password-auditing activity using available wordlists and hash inputs.

Within an incident response context, John the Ripper helps identify weak passwords, no-password conditions, and broader credential security problems that may enable unauthorized access.

Recommendations

- Use Wireshark during incident triage to inspect suspicious network communications and validate packet-level evidence.
- Use IDA Pro when malware, suspicious executables, or DLL-based payloads require deeper code-level analysis.
- Use Nikto to quickly assess whether host or web server weaknesses may have contributed to compromise.
- Use John the Ripper to test password strength and identify weak credentials that could support unauthorized access.

- Combine outputs from multiple tools rather than relying on a single platform for incident response conclusions.
- Retain screenshots and tool output as documented evidence for case review, reporting, and remediation tracking.

Conclusion

This assessment reviewed four tools in the SimSpace Virtual Range and evaluated how they support different parts of an incident response workflow. Wireshark provided strong visibility into network traffic behavior, IDA Pro supported malware and DLL analysis, Nikto enabled rapid host and web server assessment, and John the Ripper supported password auditing and credential risk evaluation.

Taken together, these tools demonstrate the value of a layered incident response toolkit. Each tool addresses a different investigative need, and together they provide useful support for detection, analysis, validation, and remediation planning within a controlled virtual environment.

Appendix A: Assessment Evidence

The following figures document key assessment steps and supporting evidence captured from the SimSpace Virtual Range during tool selection, execution, and review.

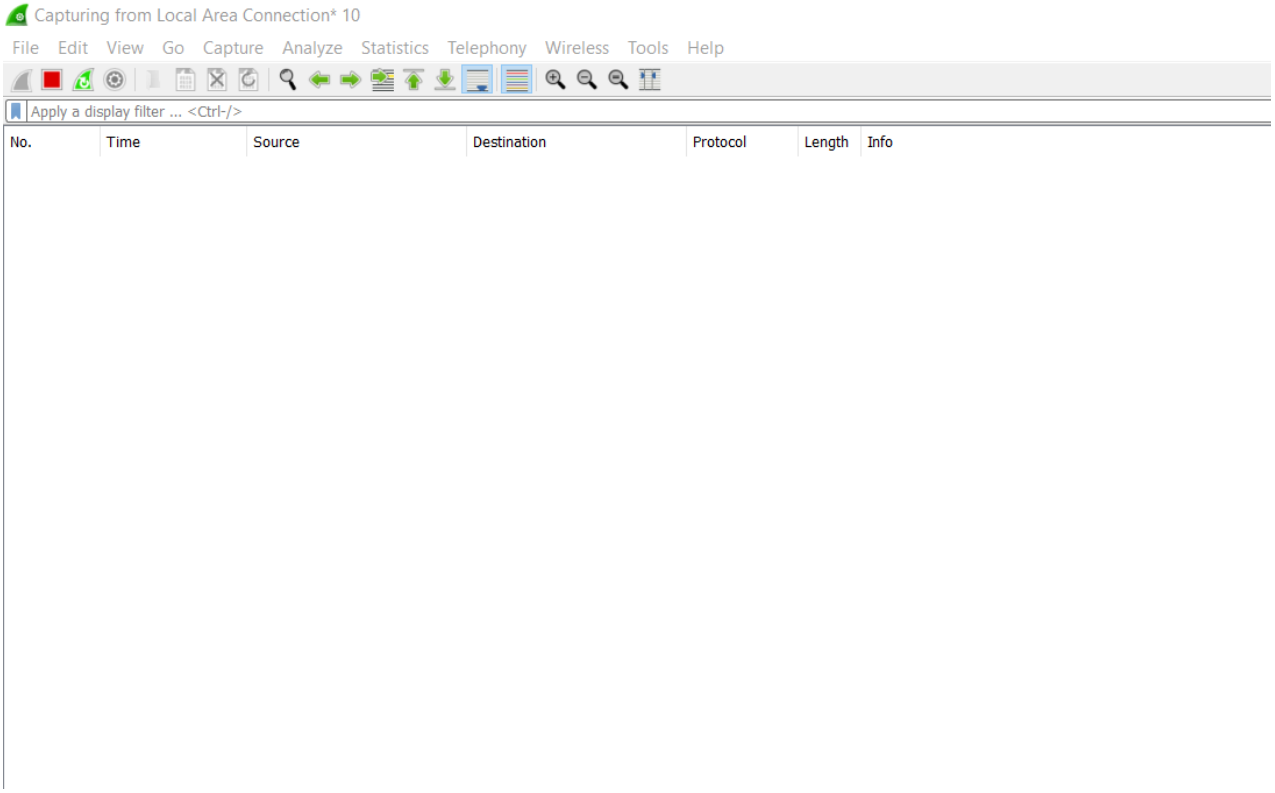

Figure A1. Kali Hunt VM application menu showing multiple assessment tools available in the SimSpace environment, including Nikto, Nmap, Sparta, and Unix privilege-checking utilities.

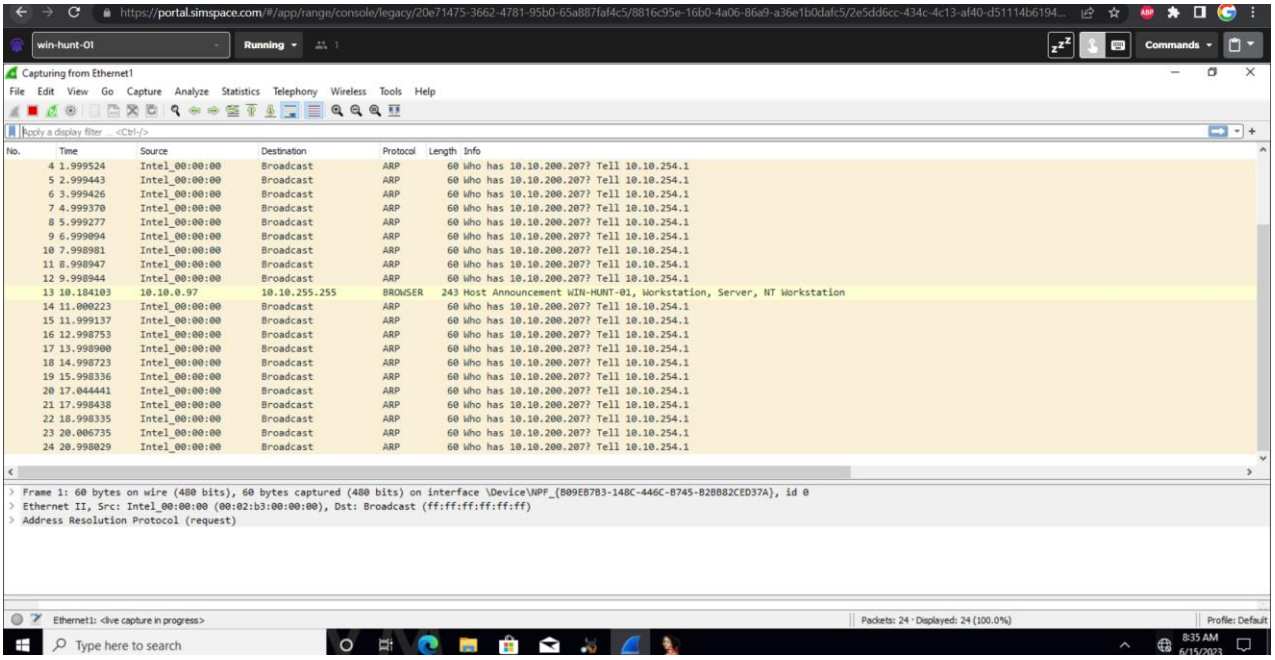

Figure A2. Sparta interface opened in Kali Hunt VM for reconnaissance and host review activities.

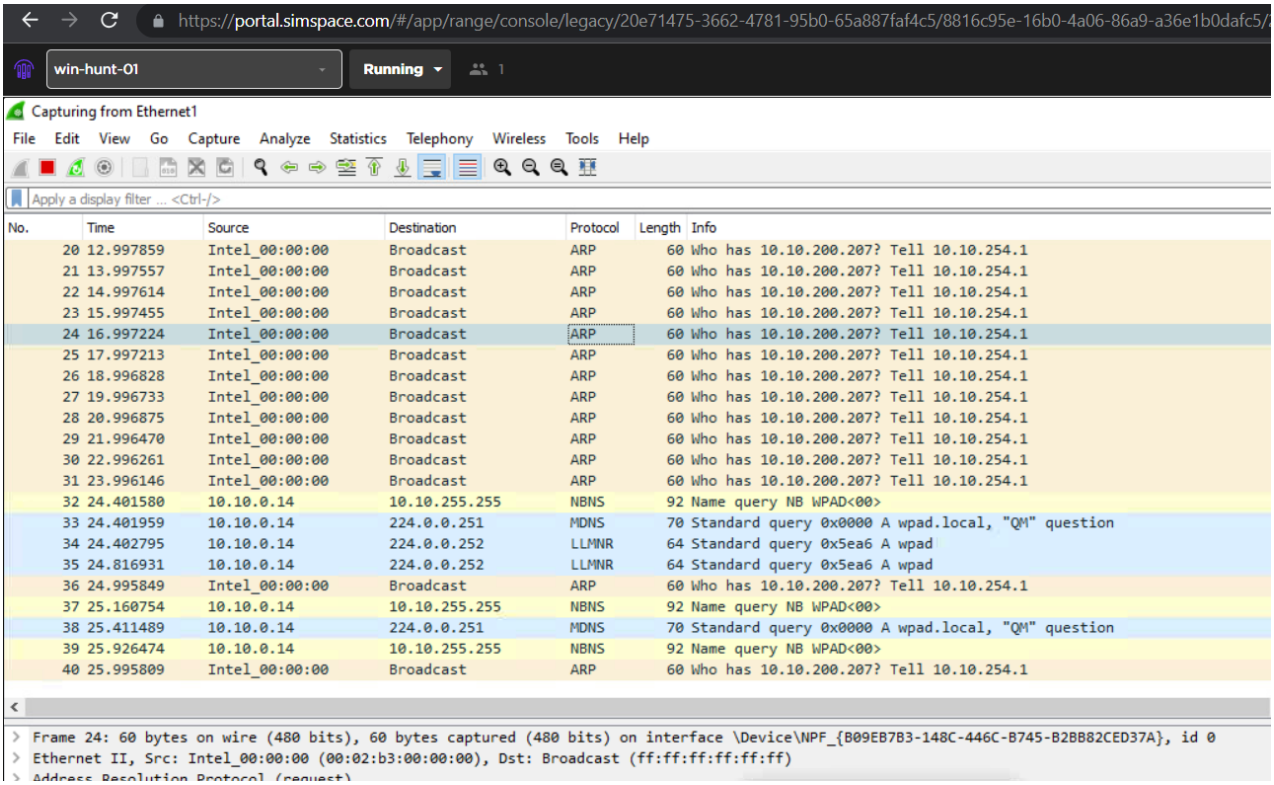

Figure A3. Windows Hunt VM desktop used to access incident response and scanning tools in the SimSpace environment.

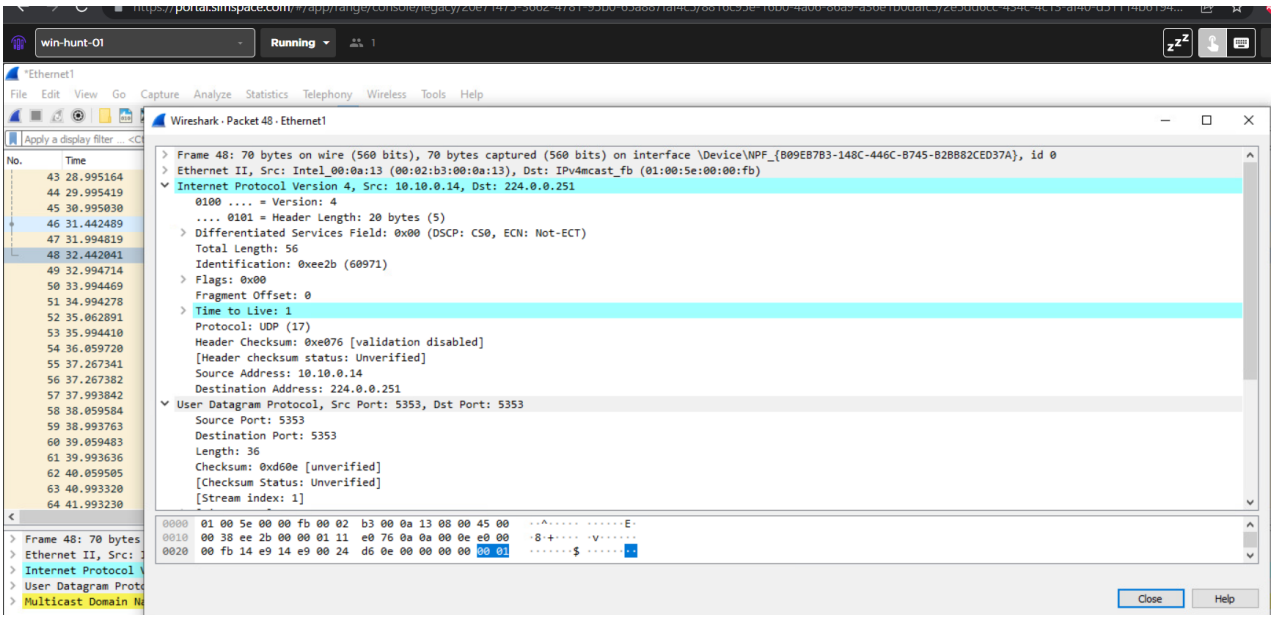

Figure A4. NetworkMiner interface available in Windows Hunt VM as an additional network analysis option.

Figure A5. Windows application search showing Nmap available for host scanning and service enumeration.

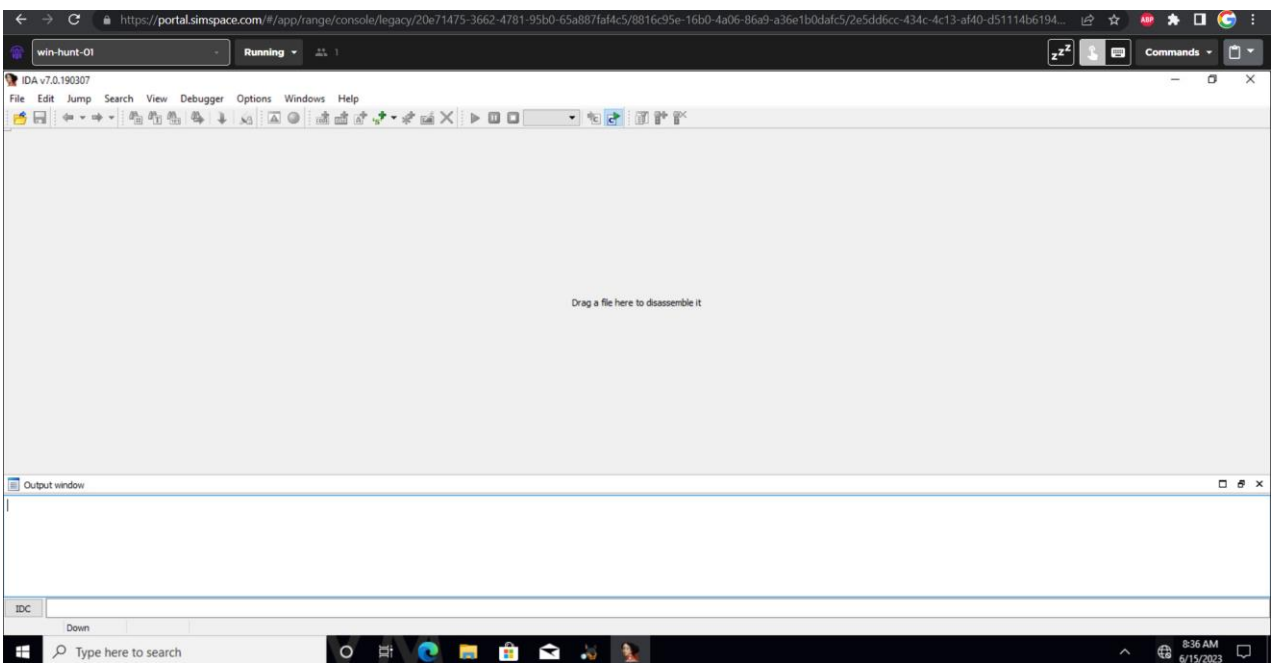

Figure A6. Sparta target-entry dialog populated for host analysis of 172.16.3.211.

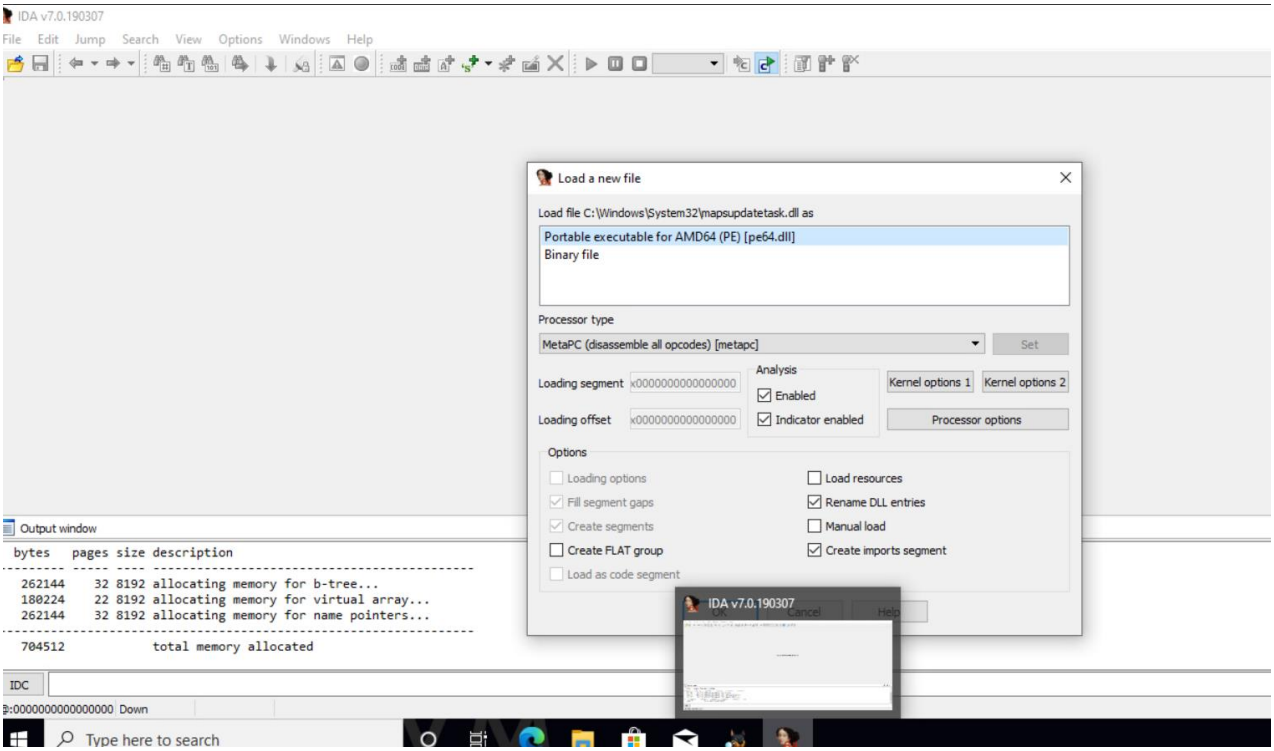

Figure A7. Sparta scan results showing discovered open ports and associated services for the target host.

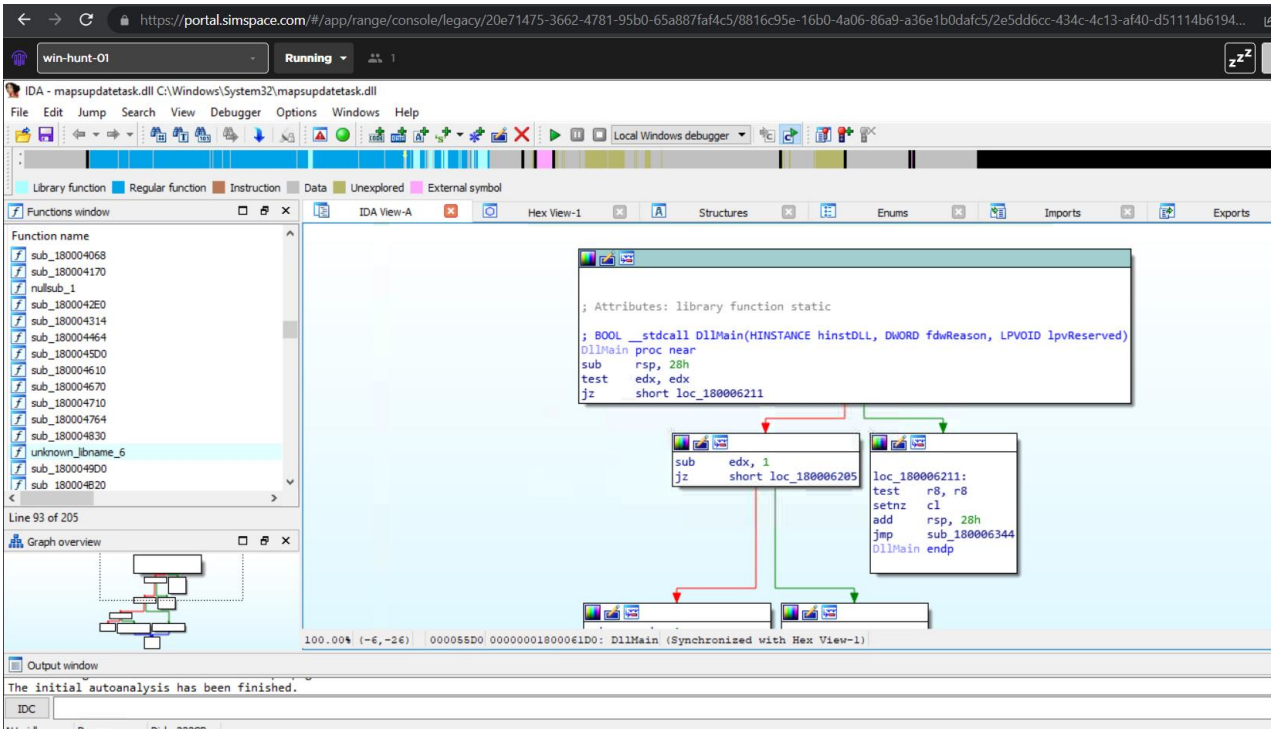

Figure A8. Sparta host summary details providing operating system, uptime, and general host information.

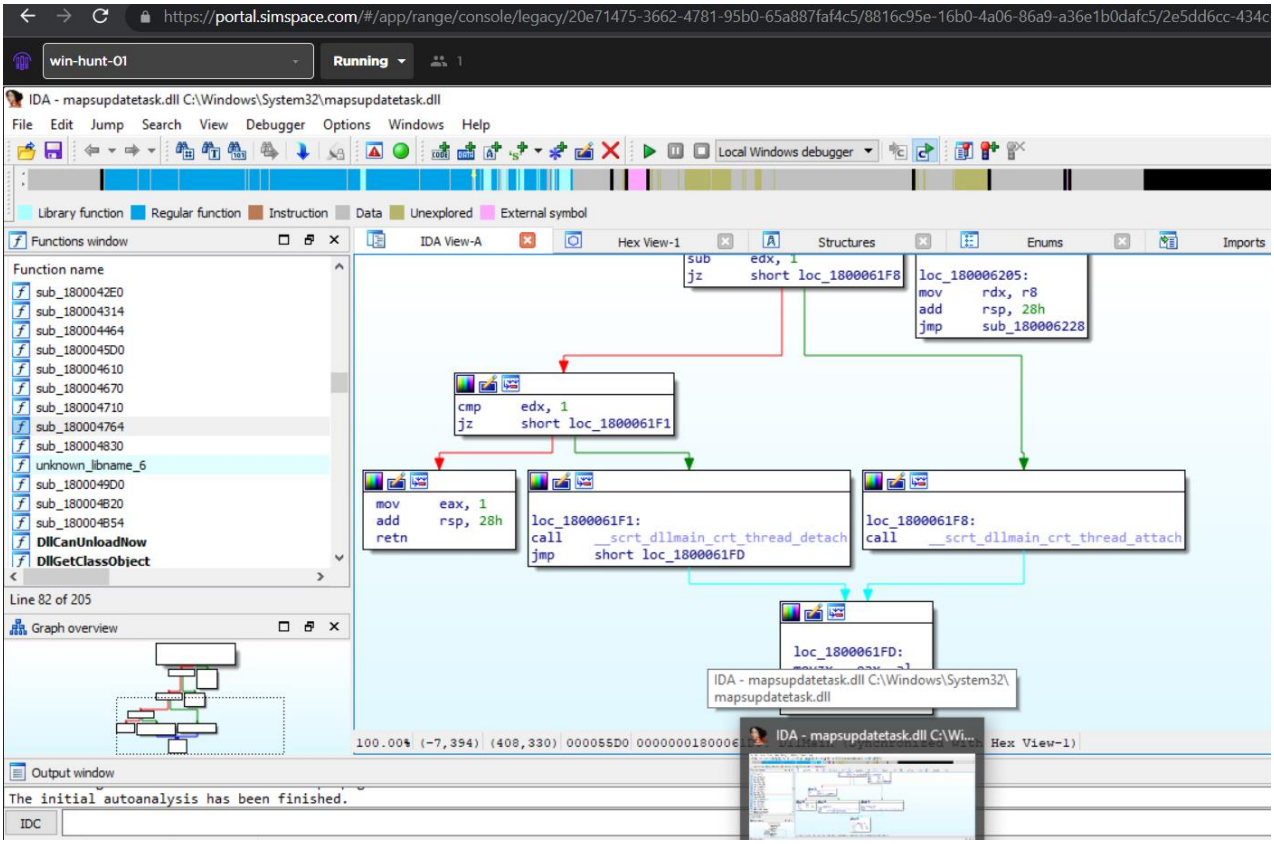

Figure A9. Sparta service details view showing protocol and service metadata returned during host analysis.

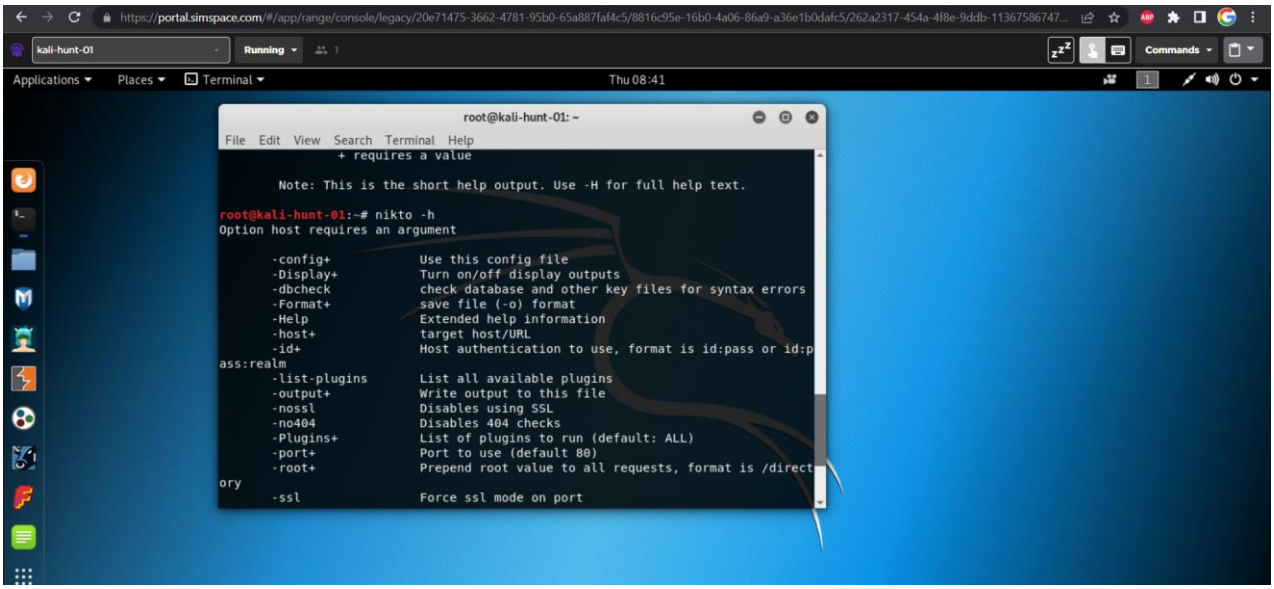

Figure A10. Nikto help and option output displayed in Kali Hunt VM, confirming tool availability and syntax.

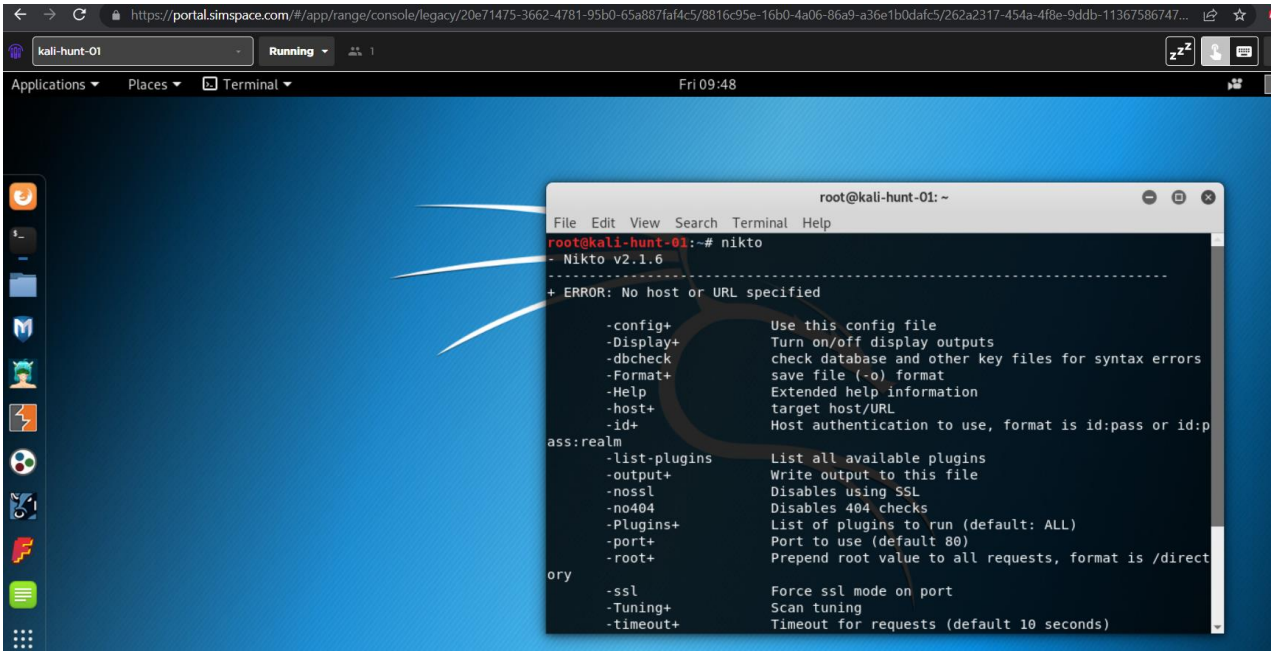

Figure A11. Nikto version output captured from Kali Hunt VM before host analysis.

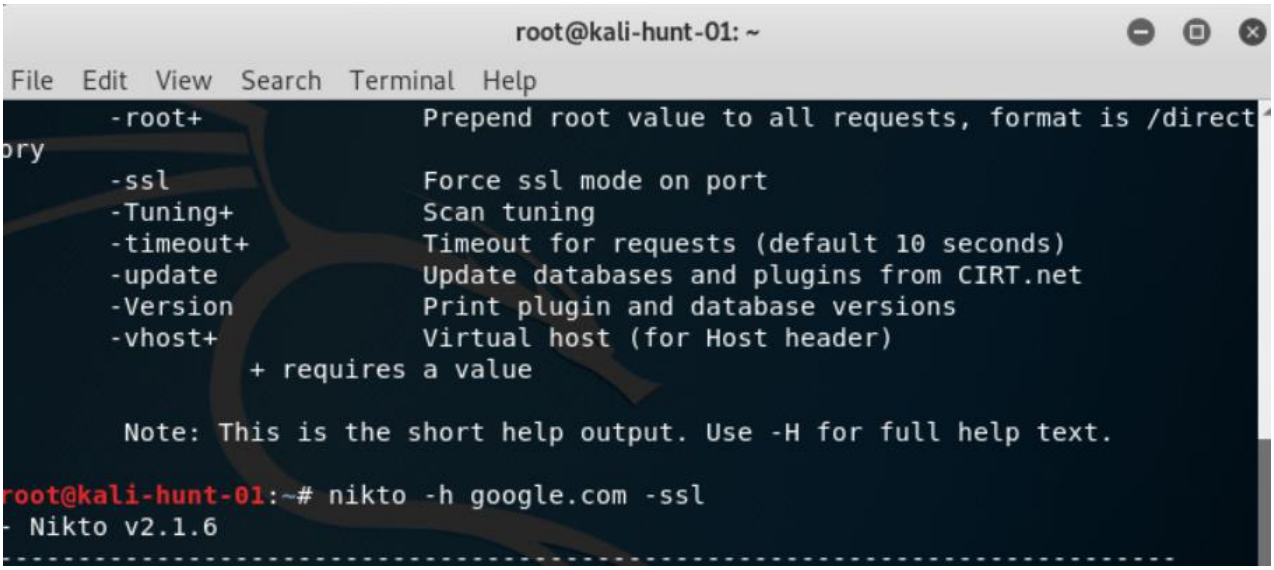

Figure A12. Nikto host-scan command prepared with SSL enabled for target validation.

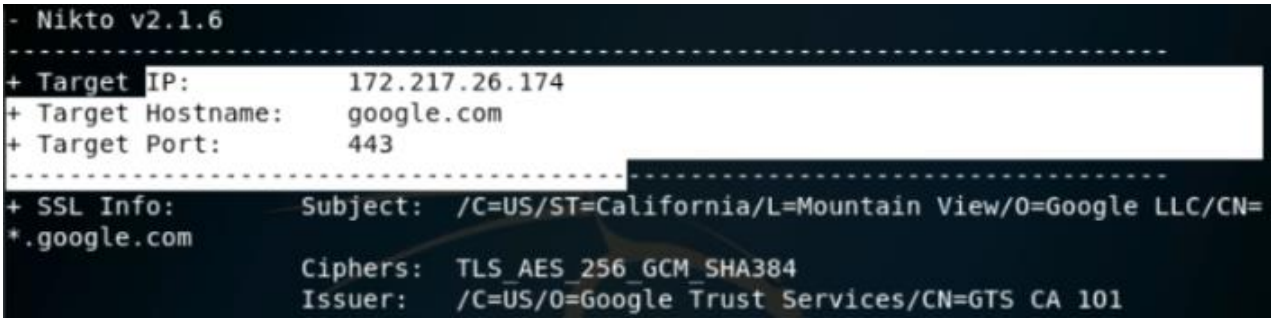

Figure A13. Nikto target information returned for google.com over HTTPS, including hostname, IP, port, and SSL certificate details.

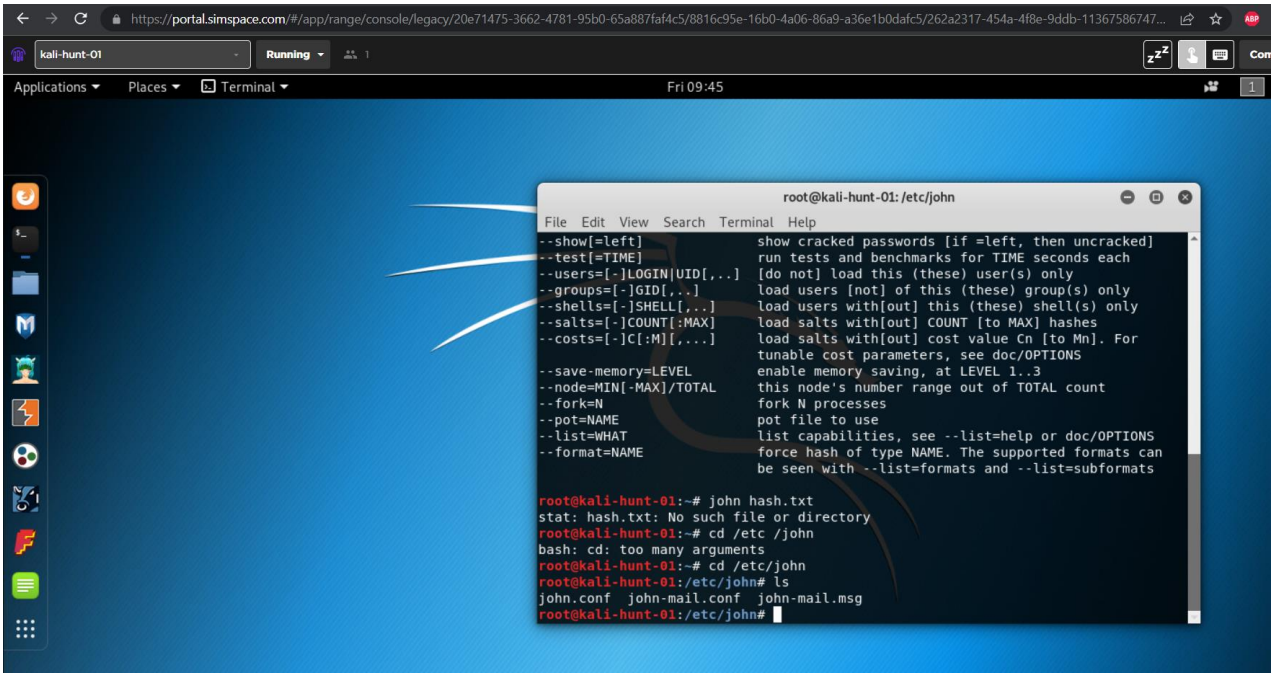

Figure A14. John the Ripper terminal interface with navigation into the John configuration directory.

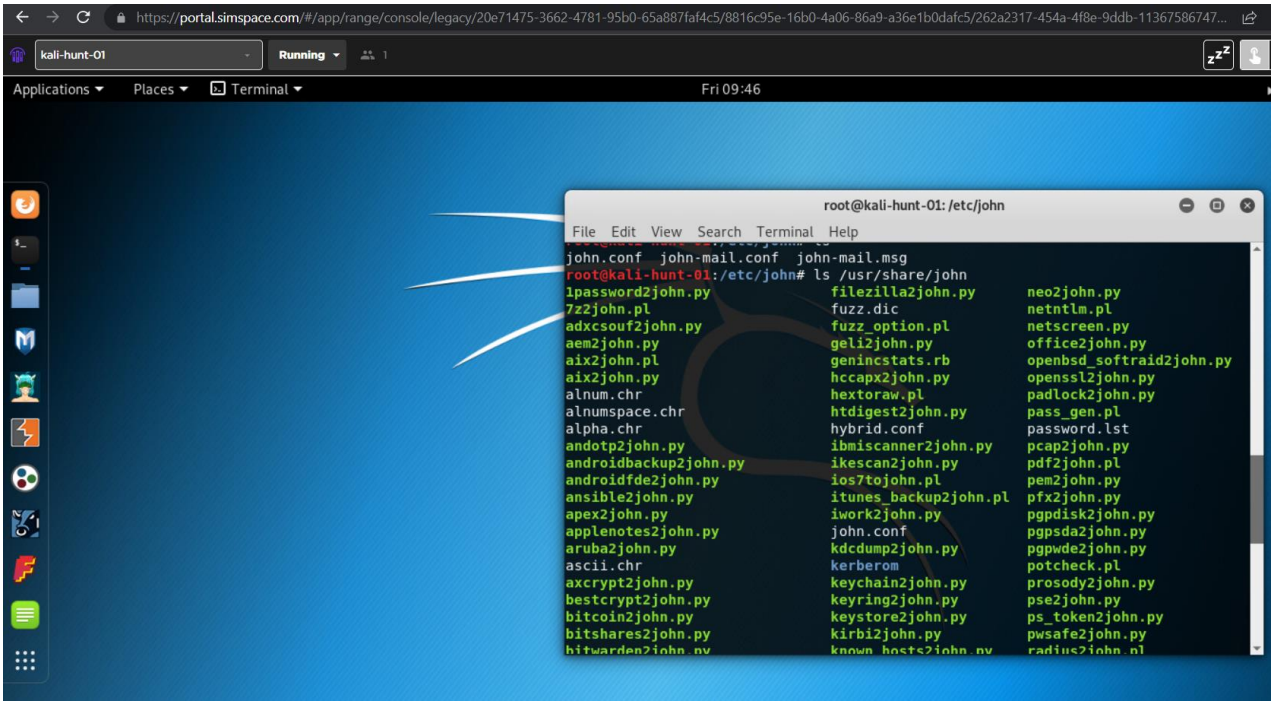

Figure A15. John directory contents displayed in Kali Hunt VM, showing password-auditing resources and scripts.

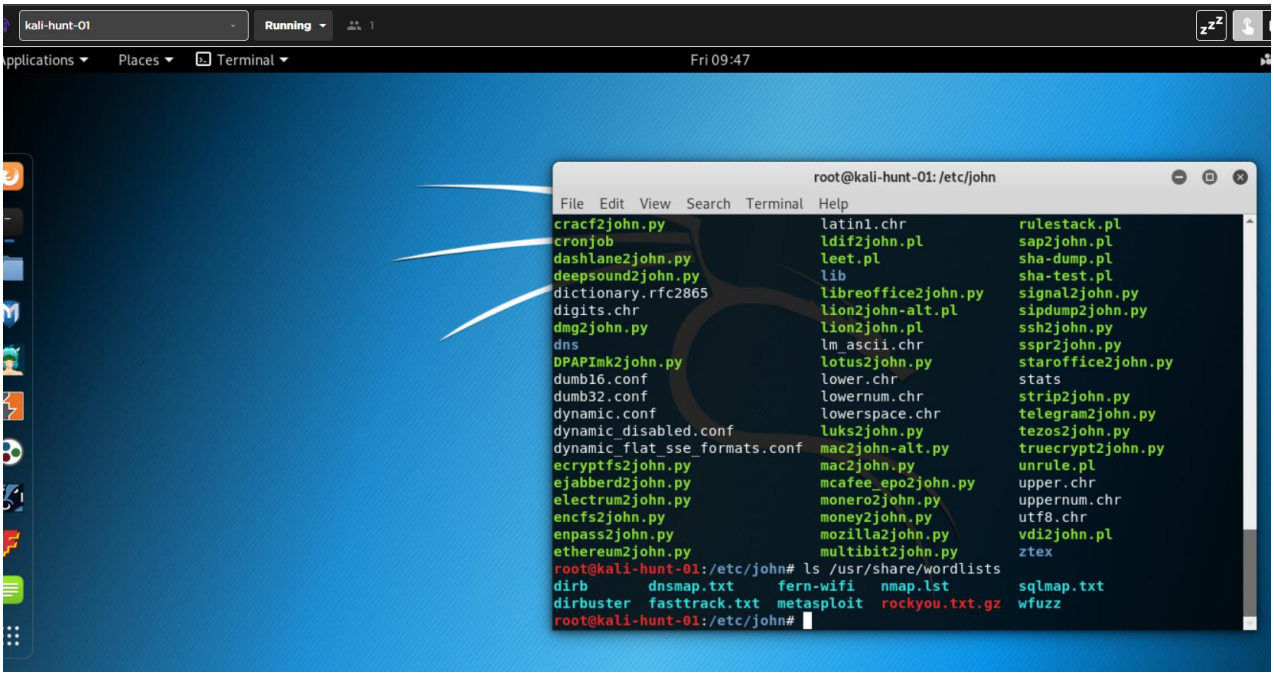

Figure A16. Wordlists available in the Kali environment for use with John the Ripper during password auditing.